Closed-loop grasp verification via structured multi-sensor reasoning under occlusion

***Abstract*—This paper presents a closed-loop execution architecture for industrial pick-and-place tasks operating under partial observability. The proposed approach employs a Behavior Tree to explicitly verify grasp and placement outcomes using multi-sensor evidence, rather than assuming action success as in open-loop execution. By reasoning over expected perceptual transitions and proximity cues, the system detects and handles grasp failures under inevitable visual occlusion. Experimental results obtained on a real robotic manipulation setup demonstrate a reduction in silent failures and an improvement in verified task completion when compared to an open-loop baseline, while maintaining identical perception and motion parameters. The results highlight the effectiveness of execution-level reasoning as a practical mechanism for handling perceptual uncertainty in industrial manipulation.**

***Index Terms*—robotic manipulation, closed-loop execution, behavior trees, grasp verification, partial observability.**

I. Introduction

Industrial robotic manipulation increasingly relies on perceptual feedback to support adaptive execution of pick-and-place tasks in complex real-world environments. In contrast to controlled laboratory conditions, industrial workspaces exhibit spatial variability, partial observability, and heterogeneous physical constraints, all of which affect the reliability and availability of perception systems. Despite these challenges, many industrial pick-and-place operations continue to rely on open-loop execution, in which motion trajectories are planned and executed under the assumption that commanded actions succeed and that perceptual feedback is available to confirm task outcomes. In practice, these assumptions are frequently violated, particularly in scenarios involving objects placed on mobile platforms or in dense configurations that lead to intentional and unavoidable occlusions during grasp execution. Consequently, visual feedback may become intermittent or unavailable during contact-rich phases of manipulation, increasing the likelihood of undetected grasp failures, execution errors, and downstream task disruptions when verification depends on single sensing modalities or instantaneous observations.

Recent research has established that perceptual uncertainty is an intrinsic aspect of robotic manipulation, rather than an exceptional circumstance. For example, Miao et al. [1] highlight the necessity of explicitly modeling occlusion and incomplete perception during task execution, as the assumption of continuous observability often fails in practice. In the domain of object localization and tracking, fiducial markers are utilized due to their implementation simplicity and detection robustness. Multi-marker configurations and multi-face fiducial objects have been shown to enhance detection rates under varying viewpoints and partial occlusions, thereby improving the reliability of object tracking in cluttered or dynamic scenes. However, these approaches do not fully resolve perception gaps that arise during critical manipulation stages, such as grasp acquisition or object placement, when visual feedback may be temporarily unavailable [2], [3].

We address the problem of verifying and recovering from grasp failures under partial observability, using only structured task execution and sensor redundancy available in typical industrial settings. Instead of adding complex, continuous perception pipelines or heavy state-estimation machinery, we adopt a pragmatic execution-level solution: encode verification and recovery directly into the task controller using a Behavior Tree (BT). The BT evaluates a small set of task-relevant conditions derived from vision, proximity sensing, and expected visibility changes, and triggers explicit recovery behaviors when verification fails. This approach shifts the burden from continuous perception to interpretable and modular decision logic, suitable for deployment on industrial systems.

This paper presents a closed-loop grasp verification approach that leverages multi-sensor redundancy and expected visibility transitions to validate grasp outcomes under occlusion. The approach is experimentally validated on a mobile-manipulation testbed, where it demonstrates improved detection and recovery of failed grasps compared to a representative open-loop baseline.

The remainder of this paper is organized as follows. Section II reviews related work on closed-loop robotic manipulation and Behavior Tree-based execution. Section III describes the experimental scenario, sensor configuration, and limitations of open-loop execution under partial observability. Section IV presents the proposed closed-loop execution architecture. Section V details the adopted experimental protocol. Section VI reports the experimental results. Section VII discusses the results in the context of the proposed approach. Finally, Section VIII concludes the paper and outlines directions for future work.

II. Related Work

Closed-loop grasping and manipulation have long been studied as a means to improve robustness in tasks where open-loop execution is insufficient. Early work such as [4] has shown that incorporating feedback into grasp execution can significantly improve task success, while highlighting that successful grasp execution cannot be assumed solely from

command execution and must instead be validated through feedback.

Recent approaches have explored closed-loop grasping in robotic manipulation using visual feedback and learning-based strategies. Methods such as [5] and [6] show that continuous visual feedback can substantially improve grasp success in cluttered environments. However, these approaches generally assume persistent visibility of the object during grasp execution, rely on frequent re-planning driven by visual observations, and uses complex, case specific neural networks that require training and validation.

To address partial observability, some works investigate reasoning over task outcomes rather than raw perception. In [7], the authors propose detecting task failures by mapping sensor observations to symbolic states, allowing systems to reason about success and failure at a higher level. While effective, such approaches typically rely on learned models trained on failure data and introduce additional complexity in terms of training, validation, and deployment.

In parallel, Behavior Trees have emerged as a framework adopted for structuring robot task execution in manipulation and industrial automation. Early works [8] demonstrate how Behavior Trees can integrate perception and action in a modular and reactive manner, enabling flexible task specification and execution. Subsequent analyses and implementations [9] [10] [11] further establish Behavior Trees as a suitable abstraction for safety-critical and explainable robotic control. Nevertheless, in most existing systems, Behavior Trees are primarily used for task sequencing, while verification of action outcomes is either implicit or handled by separate perception modules.

From a practical perception perspective, fiducial markers remain a common solution for reliable object and platform localization in controlled environments. The comparative study [12] provides a systematic evaluation of marker families and highlights their robustness and repeatability under varying conditions. In experimental manipulation setups, fiducials are often preferred when the focus is on evaluating control and execution strategies rather than advancing perception algorithms.

In contrast to approaches that rely on continuous visual servoing, learned failure models, or explicit probabilistic state estimation, the method presented in this work focuses on execution-level verification. Grasp success is assessed through interpretable condition checks derived from multiple sensors and expected visibility transitions, all embedded directly within a BT. This design emphasizes robustness under occlusion, transparency of decision logic, and ease of integration into industrial robotic systems.

III. Problem Statement and Proposed Solution

The experimental setup consists of an industrial robotic manipulator (UR10) equipped with a suction-based end-effector operating in a shared workspace with an AMR (Instor Jaguar). The manipulator is responsible for performing top-down pick operations, while the AMR serves as a mobile support platform that transports objects within the workspace.

The end-effector is a vacuum-based gripper designed for planar grasping, providing binary suction actuation, and a distance sensor mounted near the gripper to measure proximity to the nearest surface. The system is implemented using ROS 2, enabling distributed communication between perception, sensing, and execution components.

The grasping task addressed in this work involves picking an object that is placed on top of the AMR. This configuration results in intentional and unavoidable occlusions during grasp execution, particularly at the moment of contact and object lifting, which makes reliable grasp verification challenging. The task follows a structured physical sequence with distinct observability characteristics across its phases, as summarized in Table I.

A. Sensor Suite and Grasp Scenario

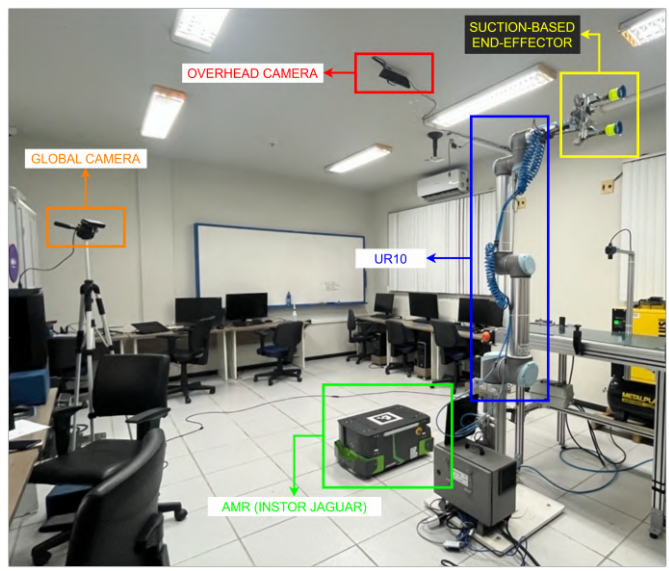

Fig. 1. Experimental setup used in the manipulation scenario. The system comprises a UR10 industrial manipulator equipped with a suction-based end-effector, an AMR (Instor Jaguar) serving as a mobile support platform, and two cameras providing global and overhead views of the workspace.

Figure 1 illustrates the physical layout of the experimental setup, including the manipulator, the mobile platform, and the sensing devices.

The system relies on a heterogeneous sensor suite composed of two depth cameras and a proximity sensor, each providing complementary but individually limited observability. An overhead camera provides a global view of the workspace and is used to detect fiducial markers attached to both the object and the AMR, enabling reliable object localization during the approach phase. However, once the gripper engages the object, the object marker becomes fully occluded.

A secondary camera is positioned to observe the AMR platform. While it cannot reliably detect the object marker during grasping, it can detect the AMR marker after the object is lifted and spatial separation from the platform occurs.

Figure 2 illustrates the visual feedback available from the two perspectives of the camera. Although the overhead view

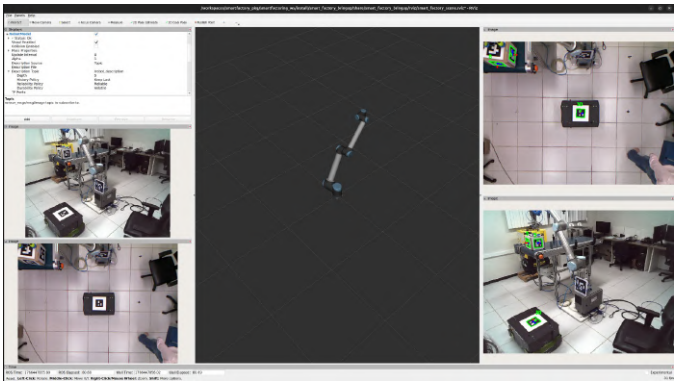

Fig. 2. RViz visualization of the visual feedback provided by the two camera perspectives. The global camera offers a complete view of the workspace, while the top-down camera primarily observes the fiducial marker on the AMR, illustrating the asymmetric and partial observability during grasp execution.

provides full scene context during the approach phase, task-relevant states related to object attachment and lift-off are not directly observable during grasp execution, resulting in asymmetric and intermittent visual feedback.

In addition to visual sensing, the ultrasonic distance sensor mounted near the gripper provides reliable contact detection during grasping. However, it cannot distinguish between an object successfully lifted and one that remains in contact with the AMR surface, making it insufficient as a standalone grasp verification signal.

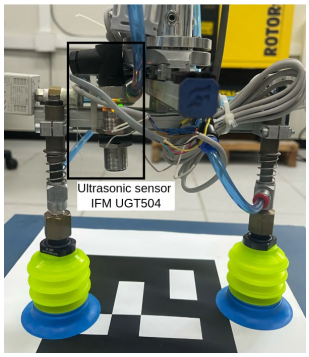

Fig. 3. Suction-based end-effector equipped with an ultrasonic proximity sensor mounted near the gripper, used to detect contact and monitor distance during grasp execution under visual occlusion.

Figure 3 shows the suction-based end-effector used in the experiments, highlighting the ultrasonic proximity sensor mounted near the gripper for contact and distance measurements during grasp execution.

As a result, each sensing modality is only partially informative, and none of them alone is sufficient to verify grasp success throughout the entire manipulation process. Grasp verification must therefore be formulated in terms of expected state transitions over time, rather than instantaneous observations from a single sensor.

Table I summarizes the observability pattern across the manipulation cycle. Instead of assuming continuous visibility,

TABLE I
SENSOR OBSERVABILITY DURING GRASP EXECUTION.

Phase	Overhead Cam	Secondary Cam	Distance
Pre-grasp	Obj visible; AMR occ.	AMR not visible	No contact
Grasp	Obj occluded	No visibility	Contact
Post-grasp	Obj may be occluded	AMR visible	Stable

the proposed architecture models the task as a sequence of phases characterized by phase-specific and transient evidence. This phase–sensor mapping enables verification based on predictable temporal transitions, exploiting the complementary nature of visual and proximity sensing.

Operationally, the grasping cycle is structured into three phases aligned with the observability pattern summarized in Table I.

In the **pre-grasp** phase, the object is supported on the AMR and must be detectable by the overhead camera to confirm target availability and estimate its pose. If the object is not detected, task execution is not initiated.

During the **grasp** phase, the manipulator approaches the object along a vertical trajectory and activates suction. Visual confirmation becomes unavailable due to marker occlusion, and verification relies primarily on contact detection provided by the ultrasonic proximity sensor.

In the **post-grasp** phase, verification depends on indirect visual evidence. The stable reappearance of the AMR marker in the secondary camera indicates that the object has been lifted and spatial separation from the platform has occurred. During this phase, grasp validity is continuously monitored until the object is released at the destination.

Figure 4 summarizes the sensing and control pipeline used in the grasping scenario. Visual observations from the two cameras are processed to estimate fiducial poses, which are then used to command the manipulator for pick and place execution. This representation highlights the implicit assumption of successful execution when grasp verification is not explicitly incorporated.

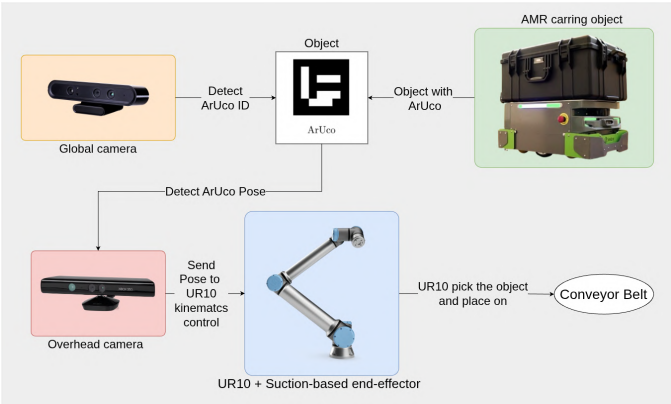

Fig. 4. Block diagram of the sensing and control pipeline in the grasping scenario. Visual observations are used to estimate fiducial poses and command the manipulator, while execution implicitly assumes grasp success in the absence of explicit verification.

B. Limitations of Open-Loop Grasp

Conventional open-loop grasp execution assumes that a grasp action succeeds once commanded, or that success can be verified through immediate post-grasp perception. In the considered scenario, these assumptions do not hold.

Due to visual occlusion during grasping and ambiguity in proximity sensing, a failed grasp may remain undetected if execution proceeds without verification. This can lead to error propagation into subsequent task stages, potentially causing collisions, dropped objects, or task failure.

These characteristics motivate the use of a closed-loop execution strategy that explicitly accounts for partial observability and leverages sensor redundancy to verify grasp success during task execution.

IV. Closed-Loop Execution Architecture

The execution architecture proposed in this work is designed to operate under partial and non-continuous observability. In contrast to open-loop approaches, task execution is not driven solely by motion commands, but by the explicit verification of whether the expected effects of each action have occurred in the environment.

Task execution is governed by a BT that integrates sensor-based condition checks at key stages of the manipulation process. Actions such as approaching, grasping, transporting, and releasing an object are only allowed to progress when their corresponding outcome conditions are satisfied. This design enables the system to distinguish between the execution of an action and the confirmation of its result.

Figure 5 illustrates the structure of the BT governing grasp execution, verification, and recovery. The tree organizes action execution, sensor-based condition checks, and recovery behaviors to handle ambiguous or failed grasps under partial observability.

A. Grasp Verification and Recovery

Grasp execution is structured as a sequence of actions followed by explicit verification. Prior to grasping, the system validates that sufficient preconditions are met, such as the availability of a reliable object pose and its presence within the expected source region.

During grasp execution, direct visual confirmation of object attachment is unavailable due to occlusion by the gripper. Grasp verification therefore relies on indirect evidence, including proximity sensing and expected changes in visual observability. Furthermore, a fixed marker attached to the AMR is visible only when there is no object over it, functioning as another step in the verification of successful grasp and movement of the object. These cues are continuously evaluated, allowing the system to assess whether the post-grasp state is consistent with a successful grasp.

Verification is integrated into the execution logic through a fallback mechanism that enables reactive recovery. If grasp confirmation fails or becomes ambiguous, the BT activates recovery behaviors that may reposition the manipulator, reattempt the grasp, or terminate execution with an explicit failure state. This design prevents silent failures and enables early error handling.

B. Transport and Placement Validation

Once a grasp is verified, object transport is conditioned on the grasp remaining valid throughout motion execution. Any loss of grasp confirmation immediately triggers recovery behavior. This includes not only a valid measure for the distance sensor, but marker visibility on the AMR.

For object placement, task completion is similarly verified by confirming the presence of the object in the destination region after release. Execution therefore depends on validated environmental state changes rather than on assumed success of motion commands.

Verification and recovery are integrated directly into the execution logic, allowing the proposed architecture to support closed-loop manipulation under occlusion without requiring continuous perception or complex state estimation. The BT offers a structured and interpretable execution framework that incorporates sensor redundancy and action verification in industrial manipulation tasks.

V. Experiments

To assess the ability of the proposed execution architecture to verify manipulation outcomes under partial observability, we conducted repeated pick-and-place experiments using a UR10 industrial manipulator operating in coordination with an AMR. Task execution was governed by a BT, which evaluated sensor-based conditions at each stage of the manipulation process. A video illustrating the experimental setup and representative executions is provided as supplementary material[1].

A. Execution Protocol

At the beginning of each trial, the presence of an object on the AMR platform was determined by the visibility of a fiducial marker and evaluated by a condition node in the BT. When the platform was inferred to be empty, no grasp was attempted. Otherwise, the BT initiated a grasp sequence.

The manipulator executed a top-down approach, causing intentional occlusion of the object marker during contact. Since direct visual confirmation was unavailable at this stage, grasp verification relied on proximity sensing and expected post-lift changes in visual observability, evaluated by verification nodes in the BT.

After lifting, the BT assessed whether the observed sensor cues were consistent with a successful grasp. If verification failed or became inconsistent, a fallback branch triggered recovery behavior or interrupted execution. During transport, grasp validity was continuously monitored, and execution was interrupted upon loss of verification.

For placement, task completion was verified by the BT through the absence of proximity readings at the gripper and detection of the object marker in the destination region. The procedure was repeated under identical workspace and motion conditions for both execution modes.

[1] https://drive.google.com/file/d/1OZwzk_X3rxCqaL9_VXMqIbi8qvN4Tnj1/view?usp=sharing

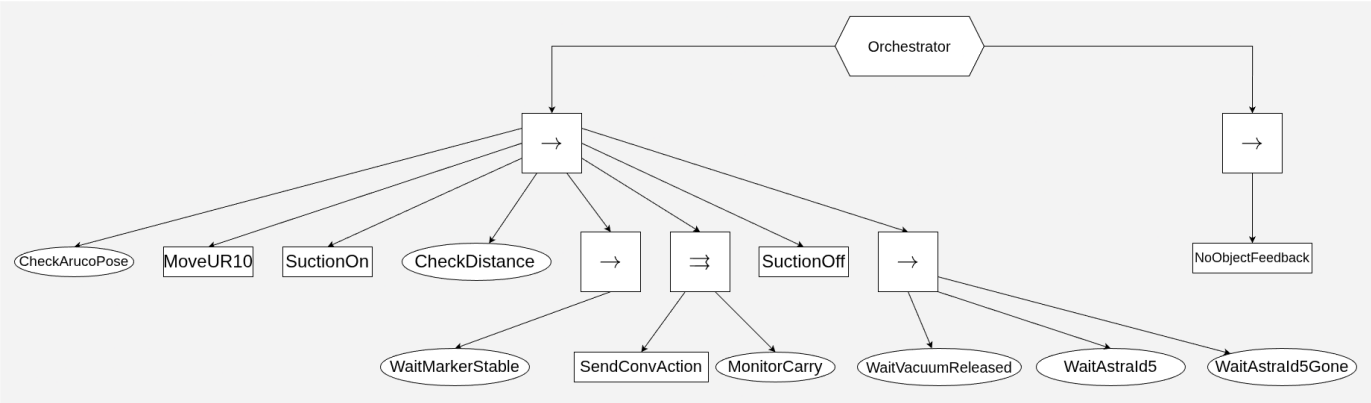

Fig. 5. BT used for closed-loop grasp execution. The tree integrates action nodes, sensor-based condition checks, and fallback structures to verify grasp success and trigger recovery behaviors under partial observability.

TABLE II
GRASP VERIFICATION PERFORMANCE OVER 15 TRIALS PER EXECUTION MODE. FAILURE DETECTION AND SILENT FAILURE RATES ARE COMPUTED OVER UNSUCCESSFUL GRASP ATTEMPTS.

Metric	Open-loop	Closed-loop
Initial grasp success rate (%)	80.0	80.0
Failure detection rate (%)	20.0	80.0
Silent failure rate (%)	80.0	20.0

TABLE III
TASK-LEVEL EXECUTION OUTCOMES OVER 15 TRIALS PER EXECUTION MODE.

Metric	Open-loop	Closed-loop
Task completion rate (%)	73.3	86.7
Interrupted executions (%)	6.7	26.7
Recovered executions (%)	0.0	13.3

VI. EXPERIMENTAL RESULTS

This section reports quantitative results comparing the proposed closed-loop execution architecture with an open-loop baseline. Fifteen pick-and-place trials were executed for each mode with identical workspace layout, motion parameters, and sensing configuration. The evaluation focuses on execution-level verification rather than perception accuracy or motion precision.

A. Grasp Verification Metrics

Grasp verification metrics reflect the system's ability to detect unsuccessful grasp attempts under partial observability. Table II summarizes these results.

Initial grasp success rates are identical across both execution modes, which is expected since perception modules, grasp poses, and actuation remain unchanged. This confirms that the proposed architecture does not improve the physical success of the grasp itself.

In contrast, substantial differences are observed in failure detection. In the open-loop baseline, most unsuccessful grasps remain undetected at execution time, resulting in a high silent failure rate. Under closed-loop execution, explicit verification embedded in the Behavior Tree allows half failed grasp attempts to be detected as they occur. Table III reports task-level execution outcomes for both execution modes.

In the open-loop baseline, undetected grasp failures often propagate to later task stages, reducing overall task completion. Since no structured recovery mechanism is available, execution typically terminates only after downstream inconsistencies occur.

In closed-loop execution, early failure detection enables structured recovery behaviors. Several trials that would otherwise fail were successfully completed after reattempted grasps, resulting in a higher verified task completion rate.

B. Verification Latency

Verification latency quantifies the additional time required to confirm grasp and placement outcomes after action execution. In the open-loop baseline, no explicit verification is performed, resulting in negligible latency, for a total average experiment time of 33.5 s. Closed-loop execution introduces an average verification latency of approximately 1.1 s for the experiments without failure, for an average 34.6 s experiment time, associated with condition evaluation and sensor-based checks embedded in the Behavior Tree.

This overhead is bounded and occurs only at discrete execution stages, representing a deliberate trade-off between execution speed and reliability.

VII. DISCUSSION

The reported results demonstrate the effect of explicit execution-level verification in manipulation tasks under partial observability. Since perception modules, motion trajectories, and grasp strategies remain identical across both execution modes, the observed differences can be attributed solely to the decision logic governing task progression.

The identical grasp success rates confirm that the proposed architecture does not affect the physical outcome of the grasp itself. This behavior is expected, as no changes are introduced at the perception or actuation levels. Instead, the contribution

of closed-loop execution lies in making grasp outcomes observable and actionable at the task level.

In the open-loop baseline, visual occlusion and sensor ambiguity frequently lead to silent failures, where unsuccessful grasps remain undetected and execution proceeds under false assumptions. These failures propagate into subsequent task stages, reducing overall task completion. The results show that such behavior is common when execution conflates action completion with outcome confirmation.

By contrast, closed-loop execution explicitly separates these concepts. Grasp success is inferred from temporally distributed and heterogeneous sensor evidence evaluated within the BT, rather than from instantaneous observations. This allows the system to detect inconsistencies between commanded actions and observed effects, even when direct visual confirmation is unavailable.

Task-level results further indicate that early failure detection enables effective recovery. Several executions that would otherwise fail are successfully completed after reattempts, increasing the verified task completion rate. Importantly, these recoveries emerge from the structure of the BT and explicit verification logic, without relying on additional perception modules or learned failure models.

The increase in verification latency reflects a deliberate trade-off between execution speed and reliability. The observed latency is bounded and occurs only at discrete execution stages, suggesting that explicit verification can be integrated into industrial workflows without significantly compromising practical deployment constraints.

Overall, the results suggest that structured execution logic plays a critical role in handling perceptual uncertainty, complementing perception systems rather than attempting to eliminate their limitations.

VIII. Conclusions

In this paper, we present a closed-loop execution architecture for industrial pick-and-place tasks operating under partial observability. The proposed approach integrates explicit outcome verification and multi-sensor evidence within a Behavior Tree, enabling task execution to be governed by verified environmental state changes rather than assumed action success.

The experimental results obtained in a real robotic manipulation setup confirm the feasibility of the proposed architecture and its effectiveness in reducing silent grasp failures. Although grasp success rates remain unchanged, closed-loop execution significantly improves failure detection and task completion by enabling early interruption and structured recovery behaviors.

The results demonstrate that reliable manipulation under occlusion can be achieved without modifying perception algorithms, motion planning, or grasp strategies. Instead, explicit reasoning at the execution level provides a practical mechanism for handling perceptual uncertainty in industrial environments.

Future work will investigate extending the proposed execution framework to more complex manipulation scenarios, longer task sequences, and multi-object settings, as well as its integration with additional sources of symbolic state and higher-level task planning.

References

[1] Y. Miao, R. Wang, and K. Bekris, "Safe, Occlusion-Aware Manipulation for Online Object Reconstruction in Confined Spaces," arXiv:2205.11719, 2022.f

[2] A. Jurado-Rodríguez, F. J. Gómez-de-Gabriel, J. L. Martínez, J. A. Ruiz-Sarmiento, J. Carrasco, and J. López-Camacho, "A Comparative Study on Planar Fiducial Markers for Robust Pose Estimation," preprint, 2023.

[3] D. García-Ruiz, P. Arnau, and J. Carnicer, "Fiducial Objects: Custom Design and Evaluation," *Sensors*, vol. 23, no. 24, p. 9649, 2023, doi:10.3390/s23249649.

[4] M. Markovic, S. Dosen, C. Cipriani, D. Popovic, and D. Farina, "Stereovision and augmented reality for closed-loop control of grasping in hand prostheses," *Journal of Neural Engineering*, vol. 11, no. 4, p. 046001, 2014.

[5] D. Morrison, P. Corke, and J. Leitner, "Closing the loop for robotic grasping: A real-time, generative grasp synthesis approach," *Robotics: Science and Systems XIV*, pp. 1–10, 2018.

[6] P. Piacenza, J. Yuan, J. Huh, and V. Isler, "VFAS-Grasp: Closed loop grasping with visual feedback and adaptive sampling," in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2024, pp. 4126–4132.

[7] P. Hegemann, T. Zechmeister, M. Grotz, K. Hitzler, and T. Asfour, "Learning symbolic failure detection for grasping and mobile manipulation tasks," in *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2022, pp. 4302–4309.

[8] C. Paxton, A. Hundt, F. Jonathan, K. Guerin, and G. D. Hager, "CoSTAR: Instructing collaborative robots with behavior trees and vision," in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2017, pp. 564–571.

[9] M. Colledanchise and L. Natale, "On the implementation of behavior trees in robotics," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 5929–5936, 2021.

[10] P. Ögren and C. I. Sprague, "Behavior trees in robot control systems," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 5, pp. 81–107, 2022.

[11] S. C. Akkaladevi, M. Propst, K. Deshpande, M. Hofmann, and A. Pichler, "Towards a behavior tree based robotic skill execution framework for human–robot collaboration in industrial assembly," in *Proc. Int. Conf. on Automation, Robotics and Applications (ICARA)*, 2024, pp. 18–22.

[12] D. Jurado-Rodriguez, R. Muñoz-Salinas, S. Garrido-Jurado, *et al.*, "Planar fiducial markers: a comparative study," *Virtual Reality*, vol. 27, pp. 1733–1749, 2023.

[13] S. Gugliermo, D. Cáceres Domínguez, M. Iannotta, T. Stoyanov, and E. Schaffernicht, "Evaluating behavior trees," *Robotics and Autonomous Systems*, vol. 178, p. 104714, 2024.

[14] E. Ghiorzi, C. Henkel, M. Klauck, M. Palmas, A. Tacchella, and A. Tacchella, "Execution Semantics of Behavior Trees in Robotics Applications," preprint, 2025.